SAS / kD=0.56 - R1 Real Local Structural Evaluation v1.0.7

Technical report and reproducibility snapshot

Author: Gonzalo Emir Durante

Freeze ID: R1_REAL_LOCAL_V107_STRUCTURAL_EVAL_PASS

Integrity manifest SHA-256: eb2899292ae72384f9eff8e8fab8e57f09b00ea0962ee822a40fbcdac2574b4c

Abstract

This report documents the R1 real local structural evaluation of SAS / kD=0.56. The snapshot validates an end-to-end real-data pipeline with paired CLEAN and HALLUCINATION candidates, full light-runner execution, corrected pair_id/source_id joins, NIG JSON serialization in v1.0.7, and validation-calibrated non-runtime structural evaluation on a held-out test split.

The best non-runtime structural composite, using Flow coherence, CRE rupture, and Negation inversion with score >= 1, achieved test F1=0.8717, precision=0.9952, recall=0.7755, and accuracy=0.8859. These results demonstrate reproducible and interpretable structural signal in SAS-light modules. They do not outperform the lexical baseline, which achieved test AUC=0.9963 and test F1=0.9968.

Dataset and runner execution

Item	Value
Source groups	2349
Total rows	4698
Validation rows	938
Test rows	3760
Full runner rows	4698 / 4698
Runner runtime	approximately 26.05 seconds
NIG conversion errors	0 / 4698

Lexical baseline boundary

Split	AUC	F1
Validation	0.9880	0.9936
Test	0.9963	0.9968

Main structural result

Composite	Test F1	Precision	Recall	Accuracy
Flow + CRE + Negation, score >= 1	0.8717	0.9952	0.7755	0.8859

Composite ablation

Rule	Test F1	Precision	Recall	Accuracy
Flow only	0.8176	1.0000	0.6915	0.8457

CRE only	0.6707	0.9927	0.5064	0.7513
Negation only	0.4777	1.0000	0.3138	0.6569
NIG only	0.1430	0.6881	0.0798	0.5218
Flow + CRE	0.8416	0.9949	0.7293	0.8628
Flow + Negation	0.8610	1.0000	0.7559	0.8779
Flow + CRE + Negation	0.8717	0.9952	0.7755	0.8859
Flow + CRE + Negation + NIG	0.8571	0.9513	0.7798	0.8699

Claim boundary

Supported: R1 real local v1.0.7 pipeline pass; NIG JSON serialization pass; Flow/CRE/Negation composite produces non-trivial non-runtime structural signal.

Not supported: superiority over lexical baselines; final hallucination detector status; broad generalization beyond this split; scientific use of runtime-derived features as structural evidence; NIG as a source-aware divergence vote in this configuration.